FLAT LOCAL MAPS INDUCING A COMPLETION ISOMORPHISM GIVE EQUIVALENCES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-equivalence-to-completion; the proof environment immediately following it is its own proof

Editorial note. The label lemma-equivalence-to-completion names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the flat local criterion: a flat local map of Noetherian local rings with m_B = m_A B and equal residue fields induces an isomorphism on completions (more-algebra-lemma-flat-unramified of More on Algebra)
- the theorem of the same chapter that the base change functor along a local map inducing an isomorphism on completions is an equivalence on the modification categories (lemma-Noetherian-local-ring)

Source reference. Original equation 0AE2: sources/stacks-restricted/pinned-original-source.tex, source lines 7444–7444. Prerequisite pointer invoked at 7545; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AE5: sources/stacks-restricted/pinned-original-source.tex, source lines 7466–7466. Prerequisite pointer invoked at 7555; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGX: sources/stacks-chow/tags.txt, source lines 13210–13210. Prerequisite pointer invoked at 7553; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $A \to B$ be a local map of local Noetherian rings such that*

(1) *$A \to B$ is flat,*
(2) *$\mathfrak{m}_B = \mathfrak{m}_A B$, and*
(3) *$\kappa(\mathfrak{m}_A) = \kappa(\mathfrak{m}_B)$*

Then the base change functor from the category (0AE2) for $(A, \mathfrak{m}_A)$ to the category (0AE2) for $(B, \mathfrak{m}_B)$ is an equivalence.

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Proof. The conditions signify that $A \to B$ induces an isomorphism on completions, see More on Algebra, Lemma 0AGX. Hence this lemma is a special case of Lemma 0AE5. □